Example 1:

$y-x+3=0$
$x^2-3xy+y^2+19=0$

$y=x-3$
$x^2-3x(x-3)+(x-3)^2+19=0$
$x^2-3x^2+9x+x^2-6x+9+19=0$
$-x^2+3x+28=0$
$x^2-3x-28=0$
$(x-7)(x+4)=0$
$x=-4$
$x=7$

$y=-7$
$y=4$

$(-4,-7)$ and $(7,4)$

Example 2:

$xy^2=4$
$xy=3$

$3y=4$

$y=\frac{4}{3}$

$x=\frac{9}{4}$

$\left(\frac{9}{4},\frac{4}{3}\right)$

Example 3:

$$\frac{x}{y}+\frac{2y}{x}=4$$

$$y=x-2$$

$$\frac{x^2+2y^2}{xy}=4$$
$$x^2+2(x-2)^2=4x(x-2)$$
$$x^2+2(x^2-4x+4)=4x^2-8x$$
$$x^2+2x^2-8x+8=4x^2-8x$$
$$x^2=8$$
$$x=\pm 2\sqrt{2}$$

$$y=-2\pm 2\sqrt{2}$$

$\left(2\sqrt{2},-2+2\sqrt{2}\right)$ and $\left(-2\sqrt{2},-2-2\sqrt{2}\right)$